Integrating coconut processing residues into sustainable food systems: technological pathways towards a circular bioeconomy

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Highlights

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A holistic 'waste-to-product' roadmap for coconut waste in sustainable food systems is presented.

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Coconut coir nanofibers create strong, biodegradable films for food packaging with UV shielding.

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Lignin-derived carbon dots enable real-time, visual freshness monitoring in smart packaging.

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Coconut meat and shell extracts offer bioactive compounds for nutraceuticals and preservatives.

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Valorizing these by-products mitigates environmental impact and enhances industry circularity.

Abstract

The coconut palm (*Cocos nucifera* L.), a globally significant tropical crop, underpins a rapidly expanding industry driven by surging demand for natural, plant-based products. With the coconut market projected to exceed USD 50 billion by 2030, sustainable management of processing residues—principally coconut shell and coir—has emerged as

a critical bottleneck for the sector's high-quality development. This review systematically evaluates the potential of these lignocellulosic by-products as feedstocks for a circular bioeconomy, focusing on their physicochemical characteristics, global processing infrastructure, and valorization pathways. We construct a novel "waste–technology–product" mapping framework that integrates established and emerging conversion technologies, including advanced bioethanol production, thermochemical conversion, and functional material synthesis. Key applications span high-value functional filtration media, sustainable construction materials, bioactive ingredients for cosmetics and nutraceuticals, and environmental remediation agents. By synthesizing mechanistic insights into recalcitrance reduction, process intensification, and multi-product biorefining, this work provides a robust scientific foundation for enhancing resource efficiency and advancing waste-to-value chains in the coconut industry.

Graphical abstract

This graphical abstract illustrates the valorization pathways of coconut waste (including coir, shell, and processing residues) into high-value products through advanced thermochemical, biological, and material synthesis technologies. Key processes include the production of bioethanol and biodiesel via enzymatic and catalytic conversion, synthesis of functional biochar and nanocomposites for environmental remediation, and development of biodegradable packaging and construction materials. The circular bioeconomy framework highlights waste-to-resource conversion, reducing environmental impact while enabling applications in energy, materials, agriculture, and biomedicine. Technological readiness and economic viability support scalable industrial implementation toward sustainable resource utilization.

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Introduction

The coconut palm is of considerable socioeconomic importance across Southeast Asia and the Pacific Islands. Botanically, the fruit is characterized by an ovoid to sub-globose shape, typically 15–25 cm in length, and is distinguished by three germ pores at the apical end. Nutritional analyses consistently identify coconut products as rich sources of water, protein, vitamins, minerals, and bioactive compounds (Roopan, 2016).These components have been associated with various health benefits, including potential roles in regulating blood glucose homeostasis, providing rapid energy replenishment, and maintaining electrolyte balance. Consequently, coconut-derived foods and beverages form a fundamental part of traditional diets throughout tropical regions. Global coconut production remains considerable, with the Food and Agriculture Organization reporting an annual yield of 62 million tonnes in 2021. Southeast Asia remains the dominant production region. Notably, the global market for coconut products is expanding significantly. Valued at USD 13 billion in 2019, projections suggest it will exceed USD 50 billion by 2030, driven by increasing demand for natural, functional foods and beverages (Warren et al., 2023).

Current processing primarily focuses on the extraction of coconut water, while coconut coir, shells, and meat residues are often discarded as waste. However, these underutilized agricultural by-products contain substantial amounts of valuable components, like cellulose and protein, offering considerable potential for resource valorization (Ismail et al., 2022; Naik et al., 2012). Fig. 1 illustrates the relationships and distinctions between coconut production, utilization pathways, and resultant waste generation. Of particular concern, coconut shells exhibit slow natural degradation cycles that can exceed a century, and accumulating coir not only occupies significant land area but also poses serious fire hazards. Consequently, the valorization of these by-products offers a dual benefit: enabling circular resource utilization while mitigating their associated environmental and safety risks.

Significant progress has been made in the valorization of coconut waste, with emerging applications ranging from the synthesis of biobased nanocomposites to the development of novel food packaging materials. Nevertheless, the existing literature lacks a systematic overview of integrated technological pathways for comprehensive coconut waste utilization. To address this critical gap, this review synthesizes current research to propose a holistic technological framework for coconut waste valorization. A tripartite ‘waste–technology–product’ resource mapping system is constructed, providing a theoretical foundation and technical reference to facilitate green, sustainable development in the coconut industry.

Resource characteristics and transformation fundamentals of coconut waste

As a commercially significant tropical crop, coconut presents substantial economic value not only through the direct utilization of its fruit components as high-value natural resources but also from its substantial yet underexploited potential for integrated valorization. A comprehensive characterization of the composition, physicochemical properties, and functional interdependencies among its lignocellulosic biomass (coir), lignified endocarp (shell), lipid-rich post-extraction residues (oil

Advanced bioethanol production from lignocellulosic coconut coir

Current research on advanced bioethanol production technology from lignocellulosic coconut coir has shifted from simple integration of traditional processes to in-depth analysis and process intensification targeting its high lignin recalcitrance. With a lignin content of 30–46% in coconut coir, the lignin-carbohydrate complex (LCC) structure constitutes a critical barrier to saccharification, necessitating pretreatment that achieves targeted deconstruction rather than general separation. Mild

Biodegradable packaging

Recent advancements in sustainable material design have highlighted the significant potential of coconut-derived biomass for developing high-performance packaging and biodegradable products. Cellulose nanofibers (CCNFs) isolated from coconut coir fiber (CCF) through peroxymonosulfate oxidation were successfully incorporated into PVA matrices to create reinforced nanocomposite films. These films show outstanding potential for transparent packaging owing to the homogeneous dispersion of CCNFs,

Process bottlenecks, industrial chain coordination barriers, and improvement methods

Coconut-derived materials hold considerable promise as sustainable alternatives in construction, particularly within a 5–10% dosage range that provides an optimal balance between mechanical strength and environmental benefit. Nevertheless, definitive substitution limits have been identified: 20% for cementitious roles and 10% as aggregates. Beyond these thresholds, strength attenuation and reduced workability occur, largely attributable to interfacial incompatibility and porous microstructures.

Conclusions and future perspectives

This comprehensive review provides the multifaceted applications of coconut-derived waste streams generated across the cultivation, processing, and consumption stages, highlighting these agricultural byproducts as valuable biorenewable resources with distinctive physicochemical characteristics that enable their transformation into high-value products through advanced conversion technologies. The current analysis highlights three principal application domains: 1) Renewable Energy Production, CRediT authorship contribution statement

Tengyue Zhang: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zhijie Gao:** Writing – original draft, Project administration, Methodology. **Dexian Zeng:** Writing – review & editing, Project administration, Methodology. **Yanhua Wang:** Visualization, Validation, Investigation, Conceptualization.

Consent to participate

All the authors in this study commented on previous versions of the manuscript. Also, they read and approved the final manuscript.

Consent for publication

All authors approved the publication of the paper.

Ethical approval

This study was of no animal or human species involved.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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